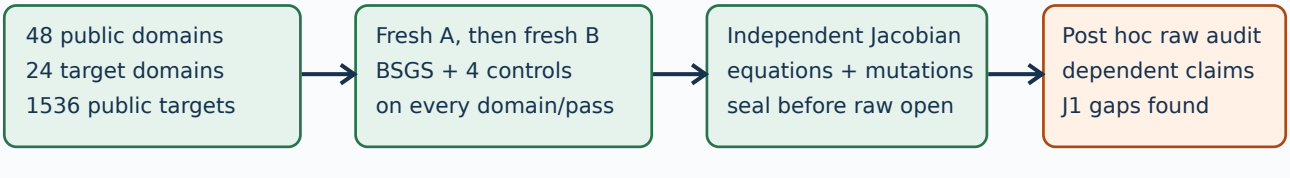
Independent static source review

J1: breaks · TASK-20261106-2ff273 · 2026-11-06

The arithmetic, complete BSGS search and independent checker fit their mathematical roles on static inspection. Two candidate-audit branches break the complete declared source contract. No solver, control, fixture, unit or scientific execution occurred. No raw candidate payload or sibling review was read.

J1 static source review: reference path and post hoc audit



F1: RECOVERED cell + null scalar + empty arrays produces no retained event.
F2: noncanonical a = r + 1 passes the residue check and can be classified valid.

Derived from source only • no solver/control execution • no historical outcome target • no performance claim

F1 — incomplete RECOVERED claim omitted

At verify.py:369–388, a correctly joined cell with status RECOVERED, words=[], scalar=null, recovered_scalars=[] creates no event. The status is only copied locally; the if-events branch omits this row from claimed cells, outcomes and disagreements. The verifier can reach its complete summary (400–408). DESIGN34–36 requires retention of every explicit claim and incomplete-claim inventory. This is a static input-shape counterexample, not an observation about the fixed raw prefix.

F2 — noncanonical multiplier can be valid

At verify.py:338–342, a,b need only be integers with nonzero residues. On E/F17: $y^2=x^3+2x+2$, G=(5,1), r=19, Q=2G=(6,3), use x=2, transformed k=2, a=20, b=1. Since 20 mod19=1, normalization gives 2 and both transformed sides equal 2G. The event can be valid even though DESIGN34 requires canonical nonzero a,b. The normalization algebra is correct; the missing canonical range check is the defect. No fixture was executed.

Scope and next action

Own joint J1 only. Correct these source branches and resolve the inventory/resource dependencies through a new owned source revision, immutable snapshot and independent review. No experiment admission, empirical correctness or scientific status change follows from this review.

Static checks and frozen false objects

Domain and affine law: solver.py:33–103 deterministically checks bounded primes, p>3, canonical coefficients/coordinates, nonsingularity, nonidentity G and unreduced rG. Identity/inverse/y=0 branches precede inversion; extended Euclid and add/double agree with SEC1 §2.2.1.

Complete BSGS: solver.py:106–131 has m=isqrt(r−1)+1=ceil(sqrt(r)). For every 0≤x<r, x=im+j with i,j<m. Babies j=0…m−1 and giants i=0…m−1 cover the search. Returns require x<r and the direct original equation. Lines164–205 build controls on all48 domains, reusing tables on24 target-bearing domains for1536 planned targets. These counts describe intended inventory, not observed successful solves. The rQ check alone is subgroup sanity, not a general proof that Q lies in the subgroup generated by G.

Independent equation checker: verify.py:55–138 implements Jacobian doubling/mixed addition, converting by Fermat inversion, and never reduces order-check scalars. It imports neither solver nor candidate arithmetic. Lines180–279 recheck all certificate vectors, canonical scalars, equations and mutations, plus controls.

Known-false objects from the pinned plan: F17 k=1 for Q=2G fails line225; k=r+2 fails223/412–415. Offcurve, noncanonical and identity bases fail constructors. Omitted/duplicate/changed-ID vectors fail216–234. A changed incorrect scalar fails its original equation. A/B changes fail261–262. A successful mutation is fatal at227–228 or333–334. These are analytical branch assessments, not recorded control outcomes.

Freshness and seal: driver.py:95–149 launches A then B as fresh Popen processes, waits, compares exact bytes and seals. Direct equations precede producer verified:true at171–178. Independent verification is written at verify279 before candidate opens at288. The current worker spawns no descendants.

Claims retained where events exist: verify318–398 audits RECOVERED words, nonnull scalar (including zero), and every recovered_scalars array element. Invalid and undefined events are retained. The transformation x=a·k/b mod r is correct for the stated canonical multipliers. F1/F2 remain material exceptions.

Additional scoped gaps

G1: driver44–46 and verify155–156 set CPU1800 independently per process, not cumulatively across the pipeline as DESIGN40 requires. Native parent enforcement was not reviewed.

G2: verify283–315/358–398 lacks all-status census and byte offsets; trace context omits arm/transformation IDs instead of explicitly checking these joins. Exact applicability to the fixed prefix was not tested.

G3: direct-child termination and grace waits do not establish hard whole-pipeline descendant/deadline guarantees; checker polling is cooperative and absent during final output serialization. G4: checker does not compare current manifest/source hashes with seal metadata or enforce original row order. Native custody checks remain necessary. None is a claim of observed overrun or tampering.

Immutable pins and references

Pinned plan: archives/TASK-20261106-924f2e/receipt.json. Independently recomputed SHA256: 382af9469b4711867c77e22d4cb1eb79d0d9713848b5ddd2fcf5abb468a2da86

Source snapshot: da0c590a872102f5c91d37d37a96753054065954. Source hashes independently match:

solver.py: d462b17ffb10772852d6024a92be71eb768b27d3d4b619cd582c868f88953031

driver.py: e0d7e4c2690f707a26edf0552e25c59327fe9b5637ee7d512c6dafdc06438124

verify.py: a19cfc02c5f95726466eb6fdef9506eb569b04e6a0ffdeb1d322bf9ff4049c8d

HAC Chapter3 §3.6.2 Algorithm3.56/Fact3.57, local primary text lines1000–1040, supports decomposition and complete coverage. Absolute source: /workspace/classical-ecdlp-endpoint-reuse/host-reference-cost-audit-20261030/root-literature/hac-chapter3.txt.

SEC1v2 §2.2.1, local primary text lines442–494, supports nonsingularity and identity/inverse/add/double. Absolute source: /workspace/classical-ecdlp-endpoint-reuse/fresh48-20261105-witness-primary-sources/sec1-v2.txt.

Both sources: provenance retrieved local primary text; verified by /root/independent_reference_source_review_20261106. All other source reads are listed completely in checks.json. Initial combined displays were truncated and relevant portions were redisplayed individually.

Independence and limits

Attestation: task_id TASK-20261106-2ff273; joints_owned [J1]; verdicts J1=breaks; read_sibling_reports=false; blind_from_respected=true. Independent source-fit session. Historical counts within DESIGN were context only, never expected outcomes or thresholds. This is explicitly post hoc source-fit review, not an outcome-blind empirical validation. Published HEAD was supplied by the dispatcher and not independently Git-inspected.

Procedure deviation: initial source display preceded display of primary passages. Primary passages were read before report writing, recommendations and final analysis. No source edit, import or scientific execution occurred. No syntax compile was repeated. Canonical graph files were outside read scope and remain unchanged. The included diagram documents source flow; it is not a performance chart.

Budget and render record

Fixed start epoch1793982438; fixed stop1793982738; maximum whole review300 seconds including report, render and inspection. No extension, reset or retry. Reviewer memory ceiling4GiB was not instrumented; no measured RSS compliance is claimed. Final actual completion and render/inspection logs are in checks.json and report.md.

Editable source is embedded in report.md. Actual renderer: installed Python Typst module. PDF is rasterized with pdftoppm and every generated page is viewed before final delivery. The final visual-inspection.json records actual inspected pages; no unperformed inspection is asserted here.